



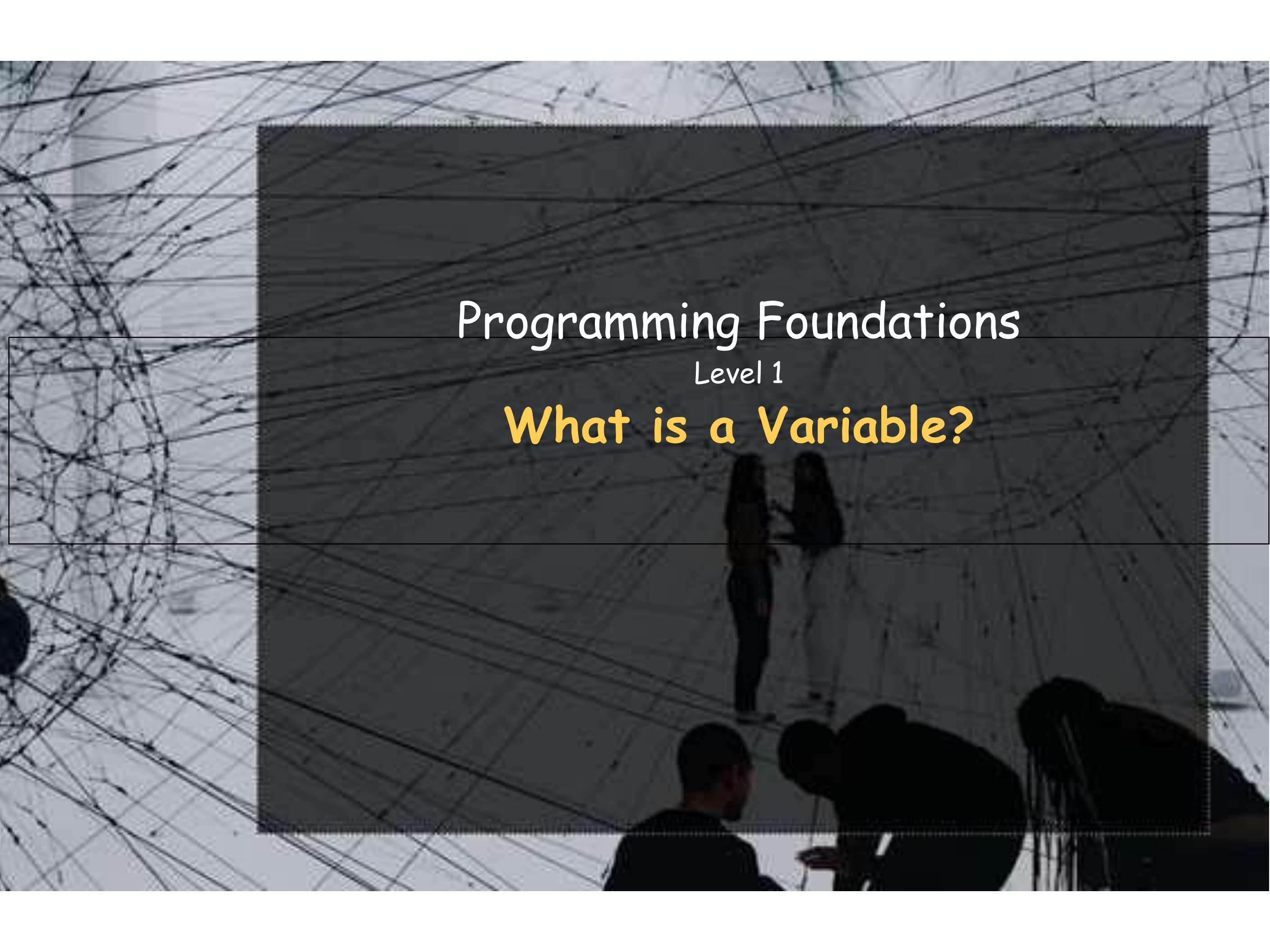
# Programming Foundations Level 1



# Programming Foundations

Level 1

## Variables & Constants: How Programs Store Data



# Programming Foundations

Level 1

## What is a Variable?

# Variables & Constants: How Programs Store Data

Programs need a place to **store information** while they run.

That place is a **variable**—a named **container** in memory that holds a value you can read and change. Sometimes you also need a **constant**—a container you **cannot** change after setting it.

In this lesson you'll see how data lives in **memory cells**, what a **memory address** is, what an **identifier** means, and how to choose the **right type** without wasting memory.



**A variable is a container for storing data values you'll reuse during program execution.**

# What are Variables?

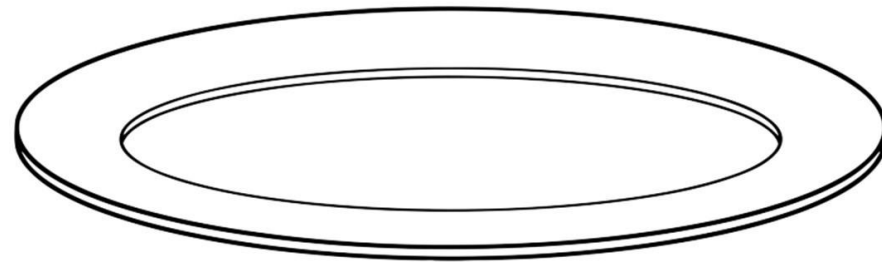
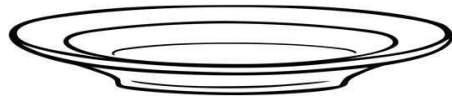
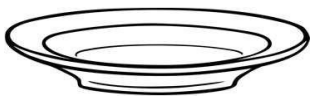
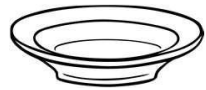
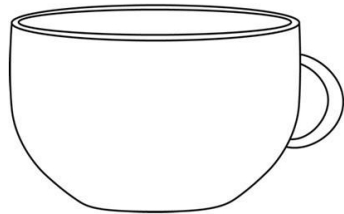
Variables are **containers** for storing data values

VARIABLES are used to store data.

**Storing information** is useful when you want **to reuse that information** many times or use it at different times while running a program.

- ◆ Think "kitchen containers": you pick the **right size** and **label** it for what it holds.

# In the kitchen



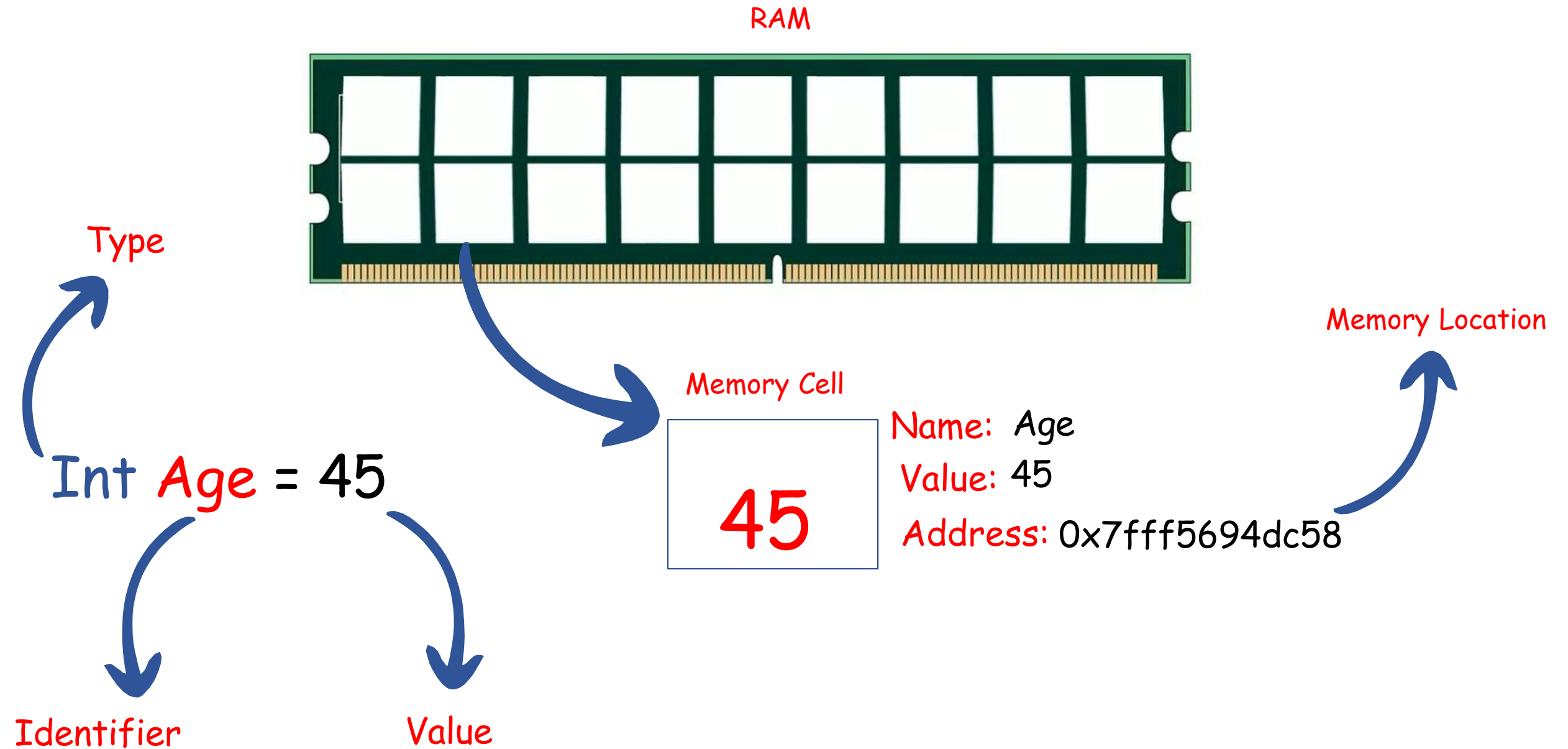
Containers



Can you use this for  
one Egg?



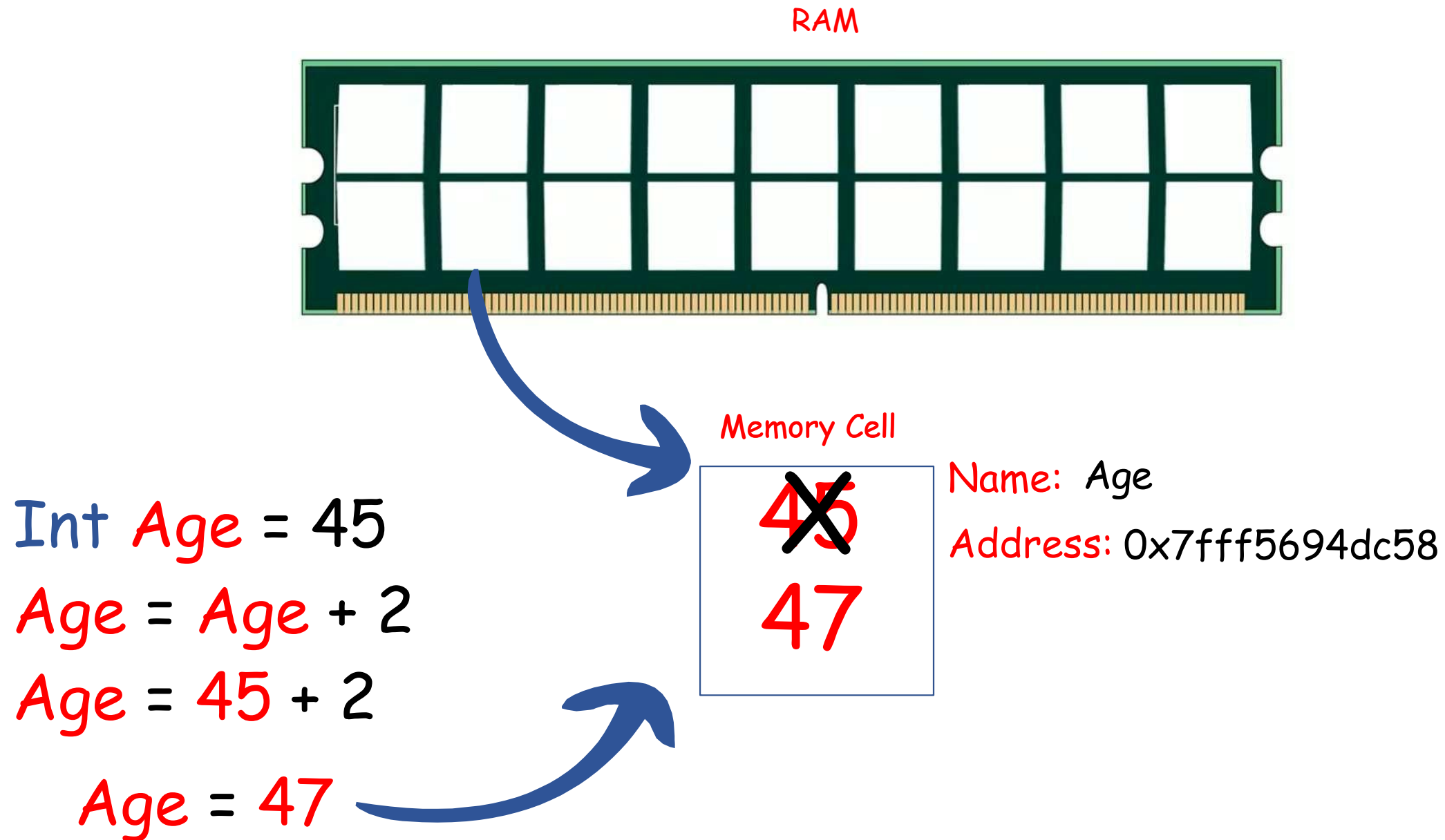
# Anatomy of a Variable (RAM view)



For a single variable you always have:

- **Address** (e.g., 0x7fff...) → Location in memory (**hexadecimal**)
- **Type** (e.g., Int, Float, String, Boolean)
- **Identifier (Name)** (e.g., Age)
- **Value** (e.g., 45)
- **Memory Cell** (the physical slot that holds the bits)

# Anatomy of a Variable (RAM view)

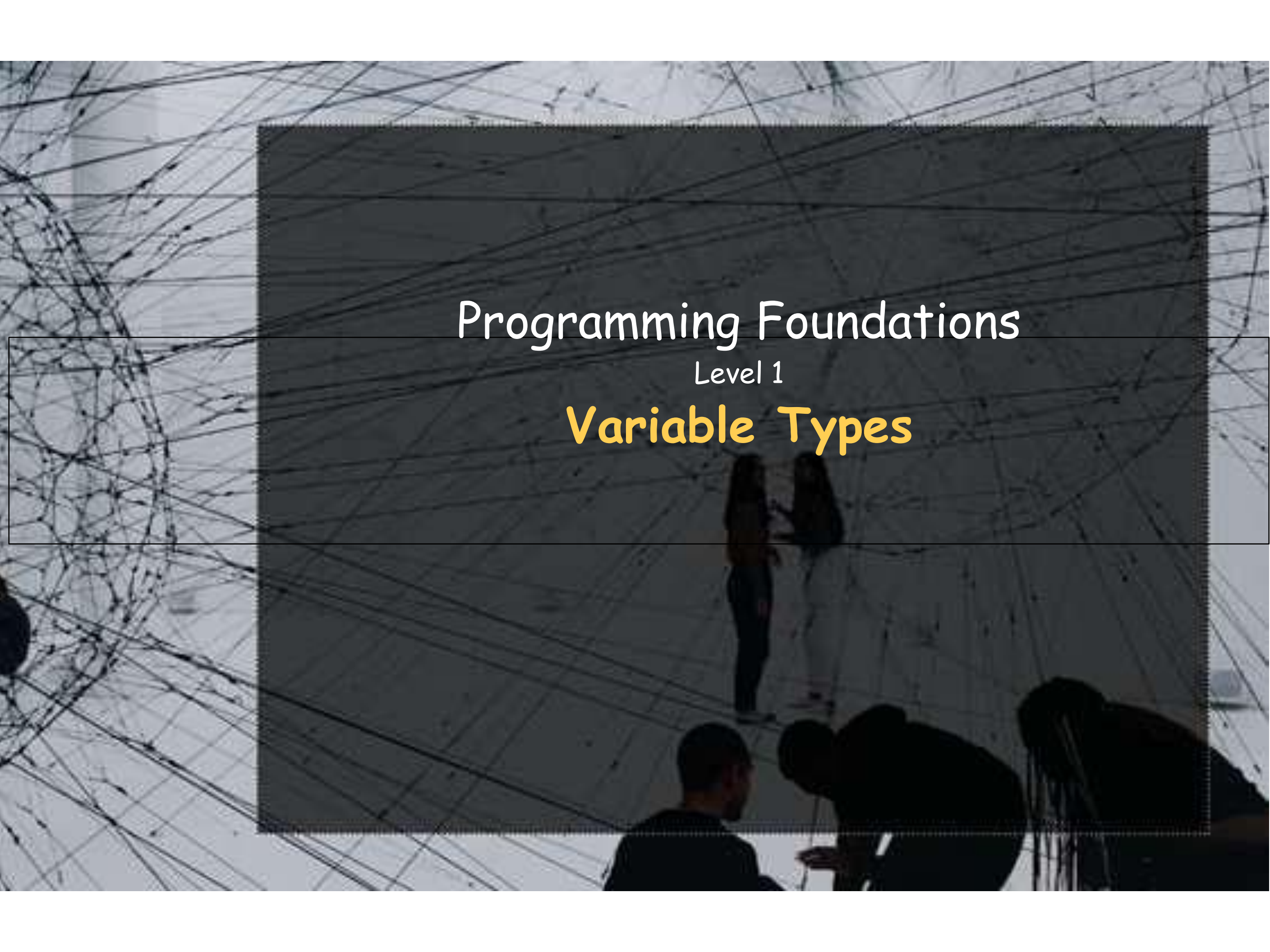


# Example :

- Type: Int
- Identifier (Name): Age
- Value: 45
- Memory cell: a slot in **RAM** that actually stores the bits
- Memory address: where this cell lives (shown in **hex**, e.g., 0x7fff5694dc58)

Updating a variable:

Age = Age + 2 → from 45 to 47. The **same cell** gets a **new value**.



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## Variable Types

# Primary Variable Types

1

## Numbers

Integers

10 , 20

Floats

5.2 , 2.3

2

## Strings

"Mohammed Abu-  
Hadhoud"

3

## Boolean

True, False

- Integer (Int): whole numbers (e.g., 0, 45, -3)
- Float/Double: numbers with fractions (e.g., 3.14)
- String: text (e.g., "Sara")
- Boolean: True/False (1/0)

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, the silhouettes of several people are visible, appearing to be in a collaborative setting, possibly a meeting or a workshop, with some individuals leaning forward as if working on a table or screen.

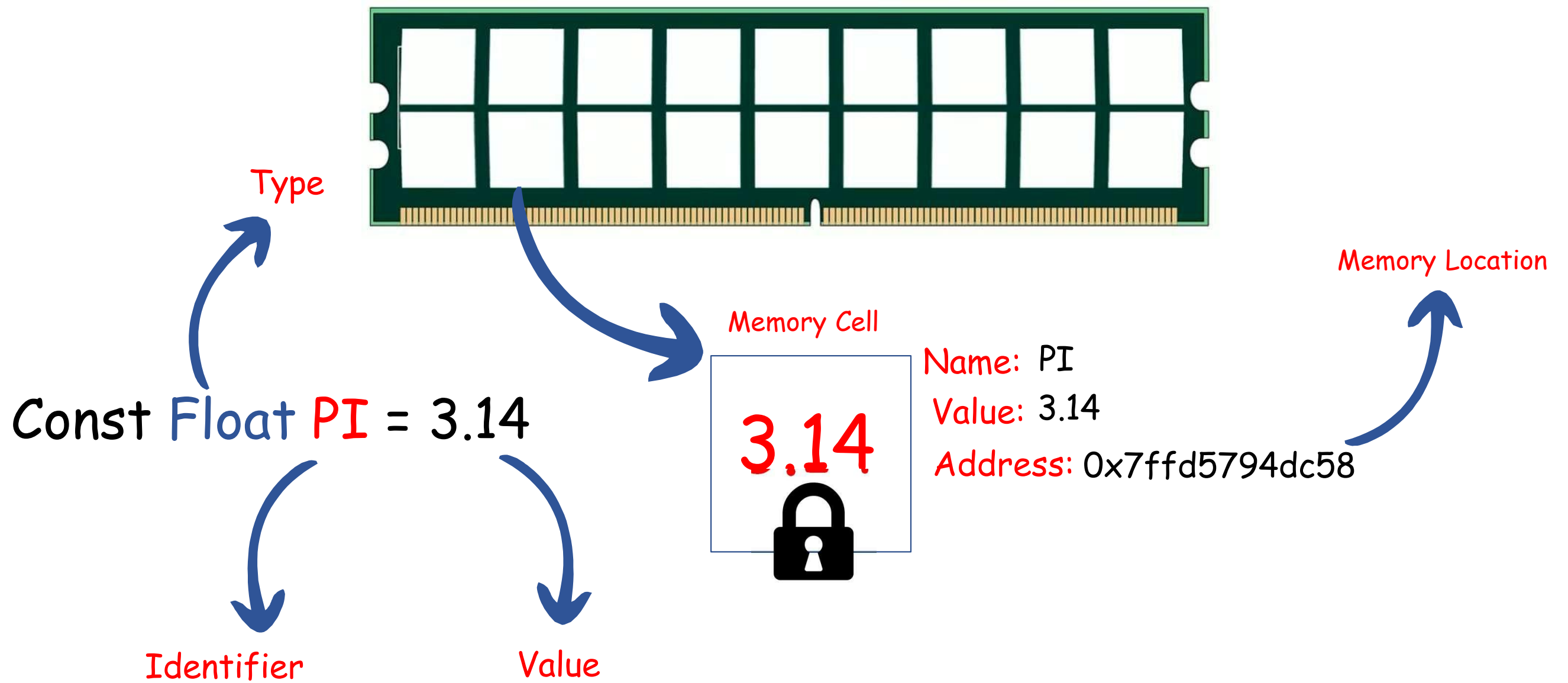
# Programming Foundations

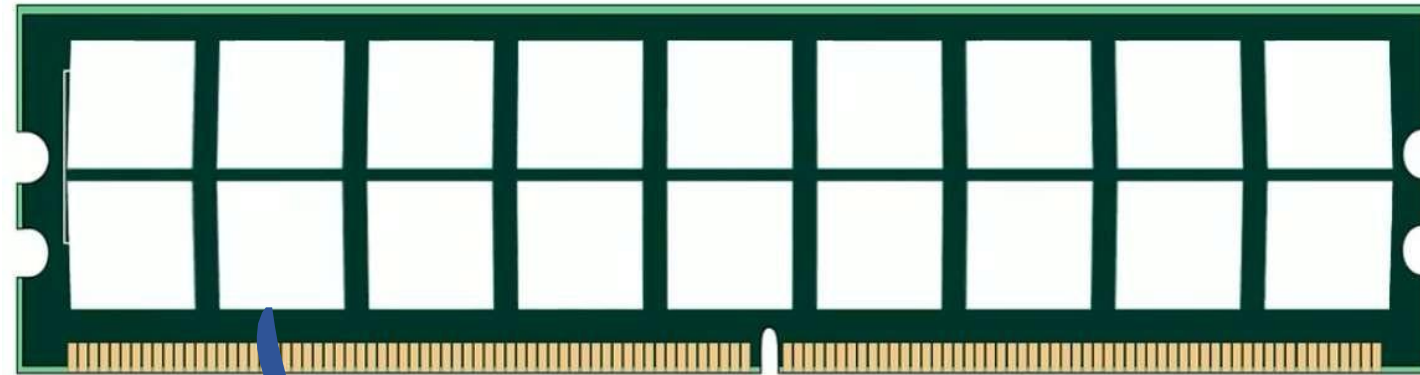
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## Constants

# What are Constants?

Constants are containers for storing data values, but they are read only.



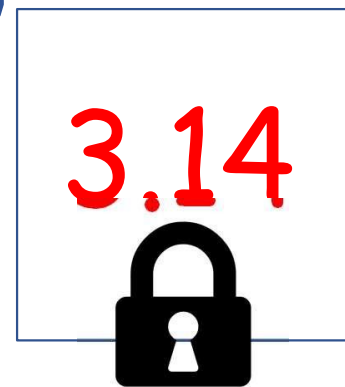


Const Float **PI** = 3.14

~~**PI** = **PI** + 2~~

**Error**

Memory Cell



Name: PI

Value: 3.14

Address: 0x7ffd5794dc58

- ◆ A constant is also a container for storing data, but read-only after initialization.
- ◆ If you try to assign a new value → **Error**.

Example: Const Float PI = 3.14 → PI = PI + 2 ❌ (not allowed).

The background of the slide is a dark, abstract image. It features a dense, intricate network of thin, light-colored lines that resemble a complex web or a neural network. In the lower portion of the image, there are dark silhouettes of several people. Two people are standing in the center, facing each other, while others are visible in the foreground, some appearing to be looking at a screen or device. The overall aesthetic is technical and modern.

# Programming Foundations

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## Performance Tips

# If you want to drink coffee, then..

**It is ok to Drink Coffee like this.**

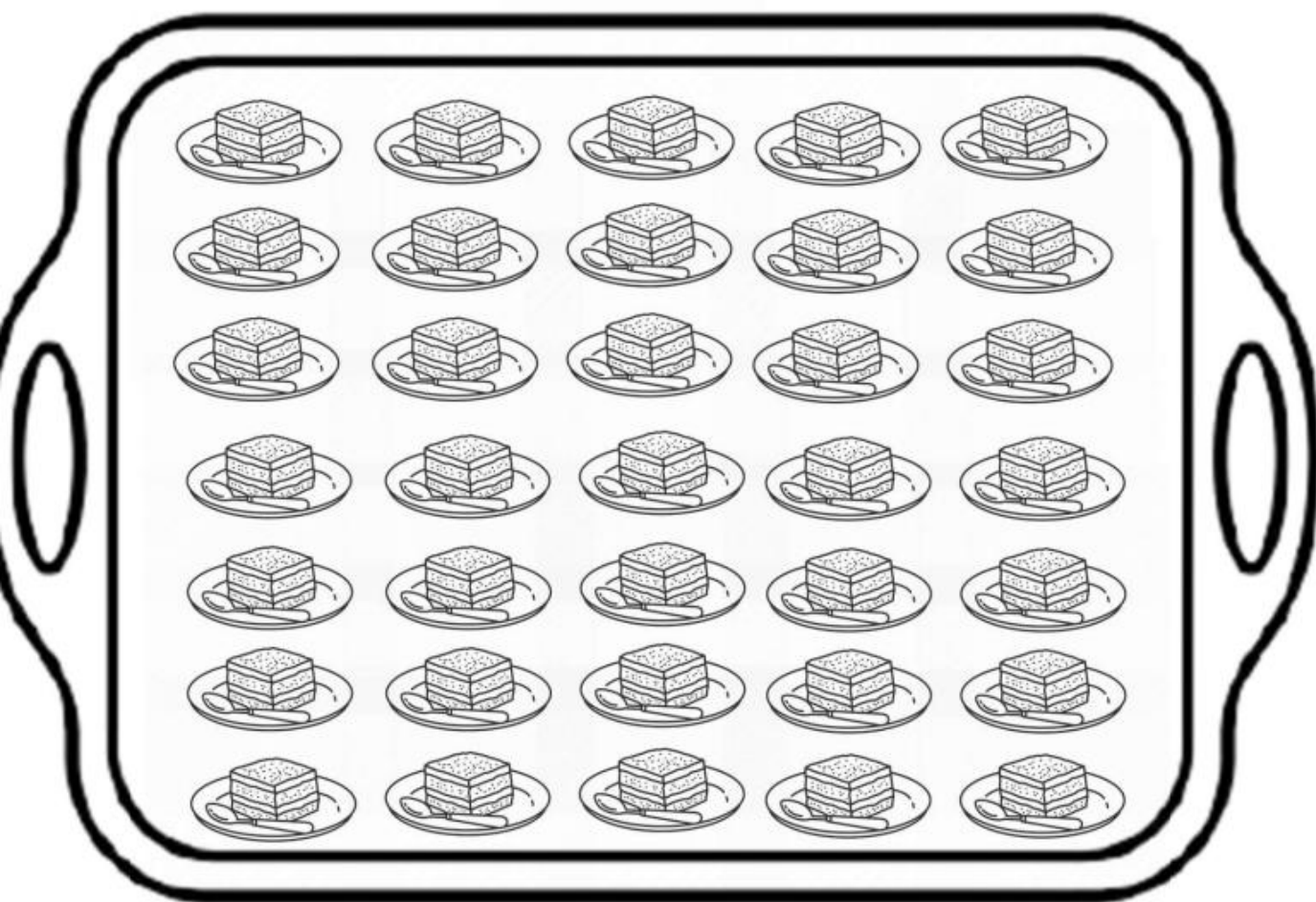


**It is not ok to drink it like this.**



35

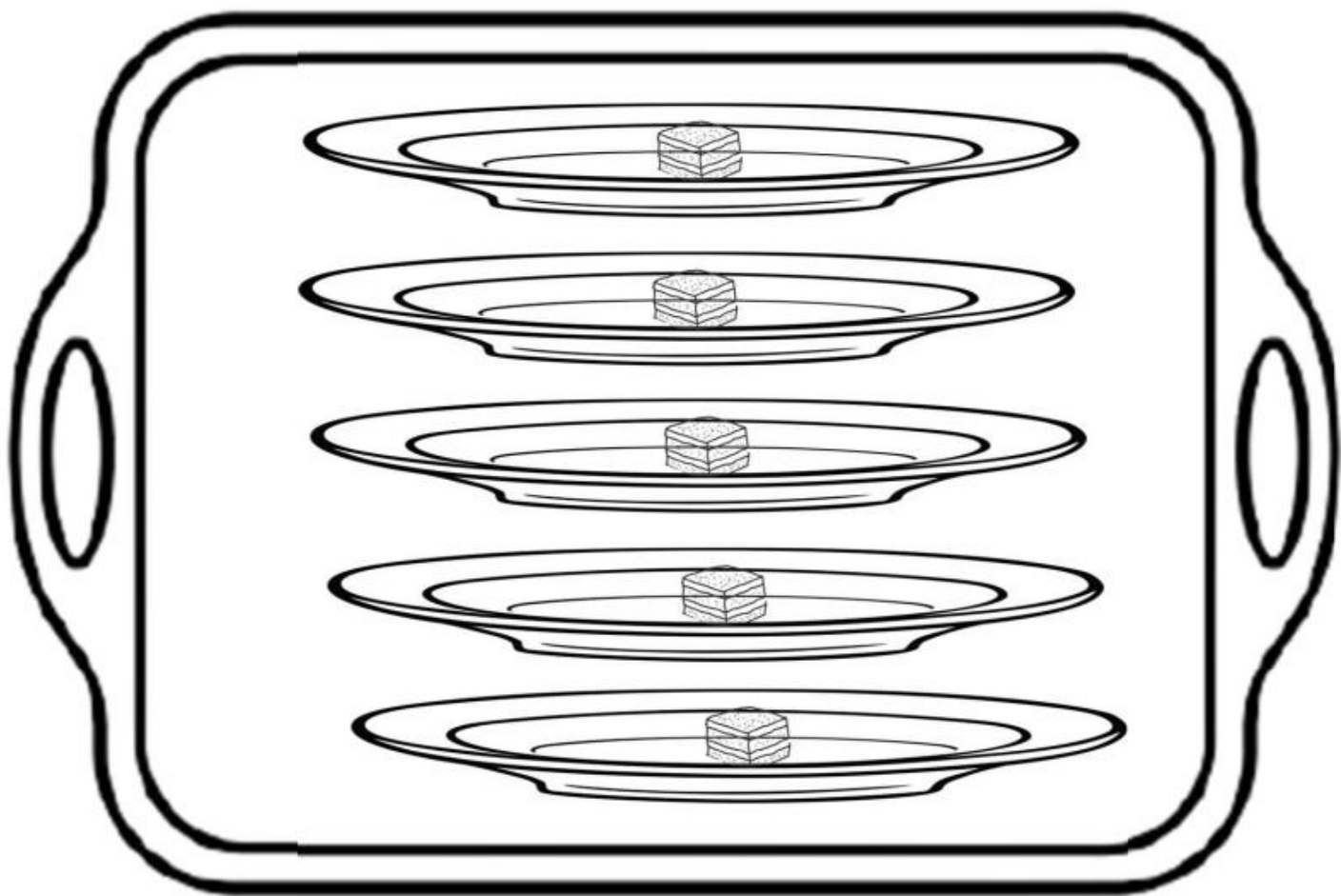
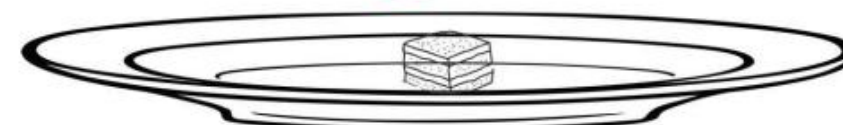
A



1 Visit

5

B



7 Visits

# Don't waste memory !!!!!

Age

Human lives 100-150 years Maximum number  
you need is 150

4,294,967,295

- Choose the **smallest sensible type** for the real-world range.
- Example : human **Age** will never be 4,294,967,295; typical max ~150 → don't reserve a huge type unnecessarily.
- Fewer/lighter variables → **fewer memory space** → better **performance**.





# Characteristics

- 1 **Variable** → **mutable** (can be changed) value in a memory cell.
- 2 **Constant** → **immutable** (cannot be changed) value; compile/runtime prevents changes.
- 3 **Identifier** → human-friendly name bound to a memory location.
- 4 **Address** → precise location in RAM (shown in **hexadecimal**).
- 5 **Type** → defines **what kind of data** and **how much memory** to use.



# Characteristics

- ◆ **Type** decides size/format → impacts **performance** and **limits**.
- ◆ **Identifier** ↔ **Address**: name you use vs where it lives in **RAM**.
- ◆ **Variable vs Constant**: mutable vs read-only; both are containers but with different rules.

By carefully choosing **names**, **types**, and **mutability**, you write code that's **clear**, **safe**, and **fast**. 🚀

# Activity

## Instructions:

Write your answers clearly and concisely.

- 1 What is a **Variable**, and why do we need it?
- 2 What is a **Constant**, and why do we need it?
- 3 What is a **Memory Cell**?
- 4 What is an **Identifier**?
- 5 What is a **Memory Address**, and what is its relation to **location** in memory?
- 6 Which **numbering system** is used to represent memory addresses?
- 7 Mention the **primary types of variables**.
- 8 What are the **number types**, and give examples for each.
- 9 What is a **String**, and give an example of it.
- 10 What is a **Boolean**, and give an example of it.



# Activity Solution



## Solution:

### ① What is a Variable, and why do we need it?

A variable is a container in memory that stores data temporarily while the program runs.

We need it to hold, update, and reuse information such as names, numbers, or results.

### ② What is a Constant, and why do we need it?

A constant is a read-only container whose value cannot change once assigned.

We need it to protect fixed values (like  $\pi = 3.14$ ) from being modified accidentally.

### ③ What is a Memory Cell?

A memory cell is a small storage unit in RAM that holds the binary bits of a variable's value.

### ④ What is an Identifier?

An identifier is the name given to a variable or constant so the programmer can refer to it easily (e.g., Age, PI).

### ⑤ What is a Memory Address, and what is its relation to location in memory?

A memory address is the exact location of a memory cell inside the computer's RAM.

Each variable is stored at a unique address — just like houses have unique addresses.

- 6** Which numbering system is used to represent memory addresses?  
Memory addresses are represented in hexadecimal (base 16) form, e.g., 0x7FFF89A2.
- 7** Mention the primary types of variables.  
Integer (Int) → whole numbers (e.g., 25)  
Float/Double → decimal numbers (e.g., 3.14)  
String → text (e.g., "Omar")  
Boolean → True/False (e.g., True)
- 8** What are the number types, and give examples for each.  
Integer (Int): Whole numbers (e.g., -10, 0, 150)  
Float (Decimal): Numbers with fractions (e.g., 3.5, 7.25)
- 9** What is a String, and give an example of it.  
A String stores text or characters, written between quotation marks.  
Example: "Hello World" or "Mohammed".
- 10** What is a Boolean, and give an example of it.  
A Boolean represents True or False values used in decisions.  
Example: IsStudent = True or HasLicense = False.

## Conclusion

- ✓ A **variable** is a named **memory cell** at an **address** that holds a **typed** value you can change.
- ✓ A **constant** is the same concept but **immutable**—great for **safety**.
- ✓ Always pick the **right type** and avoid **wasting memory**; it keeps programs **efficient and reliable**.

Thank you 😊